

$$\textcircled{1} \quad a) \quad (ABC.14)_{14} = 10 \times 14^2 + 11 \times 14^1 + 12 \times 14^0 + 1 \times 14^{-1} + 4 \times 14^{-2}$$

$$= 2126.091837$$

$$\begin{array}{r|l} 7 & 2126 \\ \hline 7 & 303 \rightarrow 5 \\ \hline 7 & 43 \rightarrow 2 \\ \hline 7 & 6 \rightarrow 1 \\ \hline & 0 \rightarrow 6 \end{array} \quad \uparrow$$

$$(6125)$$

$$.091837 \times 7 = 0.642859$$

$$.642859 \times 7 = 4.500013$$

$$.500013 \times 7 = 3.500091$$

$$.500091 \times 7 = 3.500637$$

$$.500637 \times 7 = 3.504459$$

$$(.04333)$$

$$(ABC.14)_{14} = (6125.04333...)_{14} \quad (Ans)$$

$$b) \quad (110101110111.1110101010111)_2 = (?)_{16}$$

$$\underbrace{(110101110111)}_{13} \cdot \underbrace{(111010101011100)}_{14 \quad 10 \quad 11 \quad 12}_2$$

$$= (D77.EABC)$$

2.

$$(335)_6 \times (0)_6 = (0)_6$$

$$(335)_6 \times (1)_6 = (335)_6$$

$$(335)_6 \times (2)_6 = (1114)_6$$

$$(335)_6 \times (3)_6 = (1453)_6$$

$$(335)_6 \times (4)_6 = (2232)_6$$

$$(335)_6 \times (5)_6 = (3011)_6$$

Base 6 Multiplication

$$\begin{array}{r}
 335 \overset{+6}{\overset{+6}{)}} 513231 \overset{+1}{\overset{+1}{}} (1240 \\
 \underline{335} \\
 1342 \overset{+6}{} \\
 \underline{1114} \overset{+1}{} \\
 2243 \\
 \underline{2232} \\
 111 \\
 \underline{0} \\
 111
 \end{array}$$

$$\text{Result} = (1240)_6$$

$$\text{Reminder} = (111)_6$$

$$\begin{array}{cccccc}
 3. & (1110 & 1011 & 0111 & 0111 & 1010)_{\text{Excess-5}} \\
 & \underline{14} & \underline{11} & \underline{7} & \underline{7} & \underline{10} \\
 & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\
 & 14-5 & 11-5 & 7-5 & 7-5 & 10-5 \\
 & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\
 & (9 & 6 & 2 & 2 & 5)_{10}
 \end{array}$$

$$\therefore \text{Decimal} = (96225)_{10}$$

$$4. \quad 450 - 980 = (+450) + (-980)$$

$$+450 = + \left(\overset{512}{0} \overset{256}{0} \overset{128}{0} \overset{64}{1} \overset{32}{1} \overset{16}{1} \overset{8}{0} \overset{4}{0} \overset{2}{0} \overset{1}{10} \right)_2$$

$$= (000111000010)_{2's}$$

$$-980 = - \left(\overset{512}{0} \overset{256}{0} \overset{128}{1} \overset{64}{1} \overset{32}{1} \overset{16}{1} \overset{8}{0} \overset{4}{1} \overset{2}{0} \overset{1}{00} \right)_2$$

$$= (1100000101011)_{1's}$$

$$= (1100000101\underbrace{100}_{+1})_{2's}$$

$$\begin{array}{rcl} +450 & = & 000111000010 \\ -980 & = & 110000101100 \\ \hline & & 110111101110 \end{array}$$

Here, different signs produce no overflow. $[(+ve) + (-ve) = \text{no overflow}]$

Sign bit 1 indicates result is a negative number.

$$(110111101110)_{2's}$$

$$= -(001000010001)$$

$$= - \left(\overset{512}{0} \overset{256}{0} \overset{128}{1} \overset{64}{0} \overset{32}{0} \overset{16}{0} \overset{8}{1} \overset{4}{0} \overset{2}{0} \overset{1}{0} \right)_2$$

$$= -(530)_{10}$$

$$5. \quad -1980 - 1450 = (-1980) + (-1450)$$

$$\begin{aligned}
 -1980 &= - \left(\overset{1024}{0} \overset{512}{1} \overset{256}{1} \overset{128}{1} \overset{64}{1} \overset{32}{0} \overset{16}{1} \overset{8}{1} \overset{4}{1} \overset{2}{0} \overset{1}{0} \right)_2 \\
 &= \left(1 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 1 \ 1 \right)_{1's}
 \end{aligned}$$

$$\begin{aligned}
 -1450 &= - \left(\overset{1024}{0} \overset{512}{1} \overset{256}{0} \overset{128}{1} \overset{64}{1} \overset{32}{0} \overset{16}{1} \overset{8}{1} \overset{4}{0} \overset{2}{1} \overset{1}{0} \right)_2 \\
 &= \left(1 \ 0 \ 1 \ 0 \ 0 \ 1 \ 0 \ 1 \ 0 \ 1 \ 0 \ 1 \right)_{1's}
 \end{aligned}$$

$$\begin{array}{r}
 -1980 \rightarrow 100001000011 \\
 -1450 \rightarrow 101001010101 \\
 \hline
 \begin{array}{c} \text{Carry out} \\ \text{MSB} \end{array} \quad \begin{array}{c} 1 \\ \hline 001010011000 \\ \hline \end{array} \xrightarrow{\quad} 1 \\
 \hline
 001010011001
 \end{array}$$

Here, $(-ve) + (-ve) = (+ve)$ which is not true.

$\therefore$ Overflow Detected.